

### **Assignment 1: AFRM – Batch 12**

You need to implement the following models to forecast next day VaR and Backtest the VaR on **data of your choice**.

1. Historical simulation
2. Monte Carlo simulation
3. Simple Variance based approach (backtesting using 250 days data)
4. RiskMetrics
5. GARCH model with all six distributions
6. GJR-GARCH model with all six distributions
7. EGARCH model with all six distributions

You need to prepare a Table at the end where you need to compare the %age breaches, total excess capital and total excess loss of the given models to identify the best performing model.

Deadline: 14 Nov 2025.